

32. Bank

The screenshot displays the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and a user profile for K.Dhilip. The main area is divided into three sections: Questions, Program Editor, and Input/Output.

QUESTIONS 19 / 20 completed

BANK

Write a Java program to implement a basic banking system using inheritance. Create a parent class called Account with attributes such as account number, account holder name, and balance. Then, create child classes such as SavingsAccount and CheckingAccount that inherit from the

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         SavingsAccount s = new SavingsAccount("123456789", "Jane Doe", 1.5, 1000);
6         CheckingAccount c = new CheckingAccount("987654321", "John Smith", 1000, 1000);
7         System.out.println("1. Savings Account");
8         System.out.println("2. Checking Account");
9         int choice = sc.nextInt();
10        String type = sc.next();
11        double amount = sc.nextDouble();
12        if (choice == 1) {
13            if (type.equalsIgnoreCase("deposit"))
14                s.deposit(amount);
15            else
16                s.withdraw(amount);
17        } else if (choice == 2) {
18            if (type.equalsIgnoreCase("deposit"))
19                c.deposit(amount);
20            else
21                c.withdraw(amount);
22        }
23    }
24 }
```

INPUT

```
2.
withdraw
5
```

OUTPUT

```
1. Savings Account
2. Checking Account
Transaction successful. New balance: $995.0
```

33. Employee

SIMATS Online Programming Lab
JAVA PROGRAMMING

K. Dhilip
2023-01-18 10:10

QUESTIONS 18 / 20 completed

Employee

EMPLOYEE
 Write a Java program to implement a basic employee management system using inheritance. Create a parent class called Employee with attributes such as name, id, and salary. Then, create child classes such as Manager and SalesPerson that inherit from the Employee class and

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String mName = sc.nextLine();
6         int mId = sc.nextInt();
7         double mSalary = sc.nextDouble();
8         sc.nextLine();
9         Manager m = new Manager(mName, mId, mSalary);
10        String sName = sc.nextLine();
11        int sId = sc.nextInt();
12        double sSalary = sc.nextDouble();
13        SalesPerson s = new SalesPerson(sName, sId, sSalary);
14        System.out.println("Manager Details");
15        m.display();
16        m.manageTeam();
17        System.out.println("\nSalesPerson Details");
18        s.display();
19        s.makeSales();
20    }
21 }
        
```

OUTPUT

```

Manager Details
Name : Alice
ID : 101
Salary : 50000.0
Manager is managing the team.
        
```

INPUT

```

Alice
101
50000
Bob
102
30000
        
```

34. Animal-1

SIMATS Online Programming Lab
JAVA PROGRAMMING

K. Dhilip
2023-01-18 10:10

QUESTIONS 19 / 20 completed

Animal-1

ANIMAL-1
 Write a Java program to implement a basic animal classification system using inheritance. Create a parent class called Animal with attributes such as name and habitat. Then, create child classes such as Mammal, Reptile, and Bird that inherit from the Animal class and have their own unique methods for

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String mName = sc.nextLine();
6         String mHabitat = sc.nextLine();
7         Mammal m = new Mammal(mName, mHabitat);
8         String rName = sc.nextLine();
9         String rHabitat = sc.nextLine();
10        Reptile r = new Reptile(rName, rHabitat);
11        String bName = sc.nextLine();
12        String bHabitat = sc.nextLine();
13        Bird b = new Bird(bName, bHabitat);
14        System.out.println("Mammal Details");
15        m.display();
16        m.eat();
17        m.reproduce();
18        System.out.println("\nReptile Details");
19        r.display();
20        r.eat();
21        r.reproduce();
22    }
23 }
        
```

OUTPUT

```

Mammal Details
Name : Tiger
Habitat : Forest
Mammal eats food.
Mammal gives birth.
        
```

INPUT

```

Tiger
Forest
Snake
Desert
Parrot
Trees
        
```

35. Shapes-1

SIMATS Online Programming Lab
JAVA PROGRAMMING
K.Dilip (INSTRUCTOR)

QUESTIONS 18 / 20 completed

Shapes-1

SHAPES-1
 Write a Java program to implement a basic shape hierarchy using inheritance. Create a parent class called Shape with attributes such as color and area. Then, create child classes such as Circle, Square, and Triangle that inherit from the Shape class and have their own unique methods for calculating area and ...

PROGRAM EDITOR

Java
 Run
 Save

```

1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String rColor = sc.nextLine();
7         double radius = sc.nextDouble();
8         sc.nextLine();
9         Circle c = new Circle(rColor, radius);
10        String cColor = sc.nextLine();
11        double side = sc.nextDouble();
12        sc.nextLine();
13        Square s = new Square(cColor, side);
14        String sColor = sc.nextLine();
15        double base = sc.nextDouble();
16        double height = sc.nextDouble();
17        double side1 = sc.nextDouble();
18        double side2 = sc.nextDouble();
19        Triangle t = new Triangle(sColor, base, height, side1, side2);
20        System.out.println("Circle");
21    }
22 }

```

OUTPUT

```

Circle
Color: Red
Area: 70.5
Perimeter: 31.41592653589793
Square

```

INPUT

```

Red
5
Blue
4
Green
6
4
5
5

```

36. Library

SIMATS Online Programming Lab
JAVA PROGRAMMING
K.Dilip (INSTRUCTOR)

QUESTIONS 19 / 20 completed

Library

LIBRARY
 Write a Java program to implement a basic library management system using inheritance. Create a parent class called Item with attributes such as title and author. Then, create child classes such as Book and DVD that inherit from the Item class and have their own unique attributes and methods for ...

PROGRAM EDITOR

Java
 Run
 Save

```

1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String bTitle = sc.nextLine();
7         String bAuthor = sc.nextLine();
8         int pages = sc.nextInt();
9         sc.nextLine();
10        Book b = new Book(bTitle, bAuthor, pages);
11        String dTitle = sc.nextLine();
12        String dAuthor = sc.nextLine();
13        int duration = sc.nextInt();
14        DVD d = new DVD(dTitle, dAuthor, duration);
15        System.out.println("Book Details");
16        b.display();
17        b.borrow();
18        b.returnItem();
19        System.out.println("\nDVD Details");
20        d.display();
21        d.borrow();
22        d.returnItem();
23    }
24 }

```

OUTPUT

```

Book Details
Title : Java Programming
Author : James Gosling
Book Borrowed
Book Returned

```

INPUT

```

Java Programming
James Gosling
500
Java Tutorial
Oracle
120

```

37. University



SIMATS Online Programming Lab

JAVA PROGRAMMING



QUESTIONS 16 / 20 completed

University

UNIVERSITY

Write a Java program to implement a basic university management system using inheritance. Create a parent class called Person with attributes such as name and age. Then, create child classes such as Student and Professor that inherit from the Person class and have their own unique attributes and methods for...

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String aName = sc.nextLine();
6         int aAge = sc.nextInt();
7         sc.nextLine();
8         String aCourse = sc.nextLine();
9         Student s = new Student(aName, aAge, aCourse);
10        String pName = sc.nextLine();
11        int pAge = sc.nextInt();
12        sc.nextLine();
13        String pSubject = sc.nextLine();
14        Professor p = new Professor(pName, pAge, pSubject);
15        System.out.println("Student Details");
16        s.display();
17        s.manageCoursework();
18        System.out.println("\nProfessor Details");
19        p.display();
20        p.teachClass();
21    }

```

OUTPUT

```

Student Details
Name : Alice
Age : 20
Course : Computer Science
Student is managing coursework.

```

INPUT

```

Alice
20
Computer Science
Dr. Smith
45
Java Programming

```

38. Games



SIMATS Online Programming Lab

JAVA PROGRAMMING



QUESTIONS 16 / 20 completed

Games

GAMES

Write a Java program to implement a basic game system using inheritance. Create a parent class called Game with attributes such as title and genre. Then, create child classes such as ActionGame and PuzzleGame that inherit from the Game class and have their own unique attributes and methods for...

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String aTitle = sc.nextLine();
6         String aGenre = sc.nextLine();
7         int aLevel = sc.nextInt();
8         sc.nextLine();
9         ActionGame a = new ActionGame(aTitle, aGenre, aLevel);
10        String pTitle = sc.nextLine();
11        String pGenre = sc.nextLine();
12        int pPuzzles = sc.nextInt();
13        PuzzleGame p = new PuzzleGame(pTitle, pGenre, pPuzzles);
14        System.out.println("Action Game Details");
15        a.display();
16        a.playGame();
17        System.out.println("\nPuzzle Game Details");
18        p.display();
19        p.playGame();
20    }

```

OUTPUT

```

Action Game Details
Title : Free Fire
Genre : Action
Level : 10
Playing Action Game...

```

INPUT

```

Free Fire
Action
10
Sudoku
Puzzle
25

```

39. Resturant

SIMATS Online Programming Lab
 JAVA PROGRAMMING

K. Dhillon
 10/20/2020

QUESTIONS: 10 / 30 completed

PROGRAM EDITOR

Java

Run Save

INPUT

OUTPUT

RESTAURANT
 Write a Java program to implement a basic restaurant management system using inheritance. Create a parent class called Menu with attributes such as name and price. Then, create child classes such as Appetizer and Entree that inherit from the Menu class and have their own unique attributes and methods.

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String aName = sc.nextLine();
6         double aPrice = sc.nextDouble();
7         sc.nextLine();
8         String aType = sc.nextLine();
9         Appetizer a = new Appetizer(aName, aPrice, aType);
10        String eName = sc.nextLine();
11        double ePrice = sc.nextDouble();
12        sc.nextLine();
13        String eSize = sc.nextLine();
14        Entree e = new Entree(eName, ePrice, eSize);
15        System.out.println("Appetizer Details");
16        a.display();
17        a.order();
18        a.serve();
19        System.out.println("\nEntree Details");
20        e.display();
21        e.order();
    }
  
```

Appetizer Details
 Name : Basil Bread
 Price : 120.0
 Type : Veg
 Appetizer Ordered
 Appetizer Served

Garlic Bread
 120
 Veg
 Fried Rice
 250
 Large

40. Forecasting

SIMATS Online Programming Lab
 JAVA PROGRAMMING

K. Dhillon
 10/20/2020

QUESTIONS: 10 / 30 completed

PROGRAM EDITOR

Java

Run Save

INPUT

OUTPUT

FORECASTING
 Write a Java program to implement a basic weather forecasting system using inheritance. Create a parent class called Forecast with attributes such as date and temperature. Then, create child classes such as SunnyForecast and RainyForecast that inherit from the Forecast class.

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String date1 = sc.nextLine();
6         double temp1 = sc.nextDouble();
7         sc.nextLine();
8         int hours = sc.nextInt();
9         sc.nextLine();
10        SunnyForecast s = new SunnyForecast(date1, temp1, hours);
11        String date2 = sc.nextLine();
12        double temp2 = sc.nextDouble();
13        double rain = sc.nextDouble();
14        RainyForecast r = new RainyForecast(date2, temp2, rain);
15        System.out.println("Sunny Forecast");
16        s.display();
17        s.predictWeather();
18        System.out.println("\nRainy Forecast");
19        r.display();
20        r.predictWeather();
    }
  
```

Sunny Forecast
 Date : 07-08-2026
 Temperature : 35.0 C
 Sunshine Hours : 10
 Prediction : Sunny Weather

07-08-2026
 35
 10
 08-08-2026
 20
 20

41. Rental

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 19 / 25 Completed

Rental

RENTAL

Write a Java program to implement a basic car rental system using inheritance. Create a parent class called Vehicle with attributes such as make and model. Then, create child classes such as Sedan and SUV that inherit from the Vehicle class and have their own unique attributes and methods for renting.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String make1 = sc.nextLine();
6         String model1 = sc.nextLine();
7         int seats = sc.nextInt();
8         sc.nextLine();
9         Sedan s = new Sedan(make1, model1, seats);
10        String make2 = sc.nextLine();
11        String model2 = sc.nextLine();
12        double clearance = sc.nextDouble();
13        SUV u = new SUV(make2, model2, clearance);
14        System.out.println("Sedan Details");
15        s.display();
16        s.rent();
17        s.returnCar();
18        System.out.println("\nSUV Details");
19        u.display();
20        u.rent();
21        u.returnCar();
                
```

OUTPUT

```

Sedan Details
Make : Toyota
Model : Corolla
Seats : 5
Sedan Rented
Sedan Returned
                
```

INPUT

```

Toyota
Corolla
5
Mahindra
Scorpio
220
                
```

42. Shape-2

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 19 / 25 Completed

Shape-2

SHAPE-2

Write a Java program to implement a basic shape hierarchy using polymorphism. Create a parent class called Shape with attributes such as area and perimeter. Then, create child classes such as Circle and Rectangle that inherit from the Shape class and have their own unique methods for calculating area and

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double r = sc.nextDouble();
6         double l = sc.nextDouble();
7         double w = sc.nextDouble();
8         Shape s;
9         System.out.println("Circle");
10        s = new Circle(r);
11        s.calculateArea();
12        s.calculatePerimeter();
13        System.out.println("\nRectangle");
14        s = new Rectangle(l, w);
15        s.calculateArea();
16        s.calculatePerimeter();
17    }
18 }
19 class Shape {
20     void calculateArea() {
21         System.out.println("Area");
                
```

OUTPUT

```

Circle
Area = 78.5
Perimeter = 31.400000000000002

Rectangle
Area = 40.0
                
```

INPUT

```

5
10
4
                
```

43. Draw

SIMATS Online Programming Lab
JAVA PROGRAMMING

K.Dhulip
7910400102

QUESTIONS 19 / 20 completed

PROGRAM EDITOR

INPUT
 square

DRAW
 Write a Java program to demonstrate polymorphism with interfaces. Create an interface called Drawable with a method called draw(). Then, create classes such as Circle and Square that implement the Drawable interface and have their own unique implementations of

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         Drawable d;
6         String shape = sc.nextLine();
7         if (shape.equalsIgnoreCase("circle")) {
8             d = new Circle();
9         } else {
10            d = new Square();
11        }
12
13        d.draw();
14    }
15 }
16 interface Drawable {
17     void draw();
18 }
19 class Circle implements Drawable {
20     public void draw() {
21         System.out.println("Drawing Circle");
22     }
23 }
24 class Square implements Drawable {
25     public void draw() {
26         System.out.println("Drawing Square");
27     }
28 }
    
```

OUTPUT
 Drawing Square

44. Overloading

SIMATS Online Programming Lab
JAVA PROGRAMMING

K.Dhulip
7910400102

QUESTIONS 19 / 20 completed

PROGRAM EDITOR

INPUT
 3
 10 20 30
 2
 2.5 3.5

OVERLOADING
 Write a Java program to demonstrate method overloading with variable arguments. Create a method called sum() that takes in a variable number of integers and returns their sum. Then, create an overloaded version of the sum() method that takes in a variable number of doubles and returns

```

1 import java.util.Scanner;
2 public class Main {
3     static int sum(int... n) {
4         int s = 0;
5         for (int i : n) {
6             s += i;
7         }
8         return s;
9     }
10    static double sum(double... n) {
11        double s = 0;
12        for (double i : n) {
13            s += i;
14        }
15        return s;
16    }
17    public static void main(String[] args) {
18        Scanner sc = new Scanner(System.in);
19        int n = sc.nextInt();
20        int[] a = new int[n];
21        for (int i = 0; i < n; i++) {
22            a[i] = sc.nextInt();
23        }
24    }
25 }
    
```

OUTPUT
 Sum of Integers = 60
 Sum of Double = 6.0

45. Overriding



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhilly
19216157289

QUESTIONS 19 / 20 completed

Overriding

OVERRIDING

Write a Java program to demonstrate method overriding with a simple calculator. Create a parent class called Calculator with methods such as add(), subtract(), multiply(), and divide(). Then, create a child class called ScientificCalculator that overrides the multiply() method to

PROGRAM EDITOR

java

Run

Save

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double a = sc.nextDouble();
6         double b = sc.nextDouble();
7         ScientificCalculator s = new ScientificCalculator();
8         System.out.println("Addition = " + s.add(a, b));
9         System.out.println("Subtraction = " + s.subtract(a, b));
10        System.out.println("Multiplication = " + s.multiply(a, b));
11        System.out.println("Division = " + s.divide(a, b));
12    }
13 }
14 class Calculator {
15     double add(double a, double b) {
16         return a + b;
17     }
18     double subtract(double a, double b) {
19         return a - b;
20     }
21     double multiply(double a, double b) {

```

OUTPUT

```

Addition = 9.0
Subtraction = 1.0
Multiplication = 30.0
Division = 1.25

```

INPUT

```

5
4

```

46. Person



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhilly
19216157289

QUESTIONS 19 / 20 completed

Person

PERSON

Write a Java program to create a class called Person with a constructor that takes in a name and age, and a method to print out the person's name and age.

PROGRAM EDITOR

java

Run

Save

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String name = sc.nextLine();
6         int age = sc.nextInt();
7         Person p = new Person(name, age);
8         p.display();
9     }
10 }
11 class Person {
12     String name;
13     int age;
14     Person(String name, int age) {
15         this.name = name;
16         this.age = age;
17     }
18     void display() {
19         System.out.println("Name : " + name);
20         System.out.println("Age : " + age);
21     }

```

OUTPUT

```

Name : Ram
Age : 25

```

INPUT

```

Ram
25

```

47. Rect-Const

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 18 / 20 completed

Rect-Const

RECT-CONST

Write a Java program to create a class called Rectangle with a constructor that takes in the length and width of the rectangle, and a method to calculate and return the area of the rectangle.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double length = sc.nextDouble();
6         double width = sc.nextDouble();
7         Rectangle r = new Rectangle(length, width);
8         System.out.println("Area = " + r.area());
9     }
10 }
11 class Rectangle {
12     double length;
13     double width;
14     Rectangle(double length, double width) {
15         this.length = length;
16         this.width = width;
17     }
18     double area() {
19         return length * width;
20     }
21 }

```

OUTPUT

Area = 50.0

INPUT

10
5

48. Bank-Const

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 19 / 20 completed

Bank-Const

BANK-CONST

Write a Java program to create a class called BankAccount with a constructor that takes in an account number and an initial balance, and methods to deposit and withdraw money from the account.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int accNo = sc.nextInt();
6         double balance = sc.nextDouble();
7         BankAccount b = new BankAccount(accNo, balance);
8         double deposit = sc.nextDouble();
9         b.deposit(deposit);
10        double withdraw = sc.nextDouble();
11        b.withdraw(withdraw);
12        System.out.println("Final Balance = " + b.balance);
13    }
14 }
15 class BankAccount {
16     int accNo;
17     double balance;
18     BankAccount(int accNo, double balance) {
19         this.accNo = accNo;
20         this.balance = balance;
21     }

```

OUTPUT

Final Balance = 6000.0

INPUT

1001
5000
2000
1000

49. Car-Const

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 19 / 20 completed

Car-Const

CAR-CONST
 Write a Java program to create a class called Car with a constructor that takes in the make, model, and year of the car, and a method to print out the car's make, model, and year.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String make = sc.nextLine();
6         String model = sc.nextLine();
7         int year = sc.nextInt();
8         Car c = new Car(make, model, year);
9         c.display();
10    }
11 }
12 class Car {
13     String make;
14     String model;
15     int year;
16     Car(String make, String model, int year) {
17         this.make = make;
18         this.model = model;
19         this.year = year;
20     }
21     void display() {

```

OUTPUT
 Make : Toyota
 Model : Corolla
 Year : 2022

INPUT
 Toyota
 Corolla
 2022

50. Student-Const

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 19 / 20 completed

Student-Const

STUDENT-CONST
 Write a Java program to create a class called Student with a constructor that takes in a name, ID number, and a list of grades, and methods to calculate and return the student's average grade and letter grade.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String name = sc.nextLine();
6         int id = sc.nextInt();
7         int n = sc.nextInt();
8         int[] grades = new int[n];
9         for (int i = 0; i < n; i++) {
10             grades[i] = sc.nextInt();
11         }
12         Student s = new Student(name, id, grades);
13         System.out.println("Name : " + s.name);
14         System.out.println("ID : " + s.id);
15         System.out.println("Average = " + s.average());
16         System.out.println("Letter Grade = " + s.letterGrade());
17     }
18 }
19 class Student {
20     String name;
21     int id;

```

OUTPUT
 Name : John
 ID : 101
 Average = 87.6
 Letter Grade = B

INPUT
 John
 101
 5
 90
 85
 80
 95
 88

51. Palindrome



SIMATS Online Programming Lab

JAVA PROGRAMMING


K.Dhillip

12/04/2023 10:00

QUESTIONS
11 / 25 completed

Palindrom

PALINDROME

Write a Java program to check if a given string is a palindrome or not.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         String rev = "";
7         for (int i = str.length() - 1; i >= 0; i--) {
8             rev = rev + str.charAt(i);
9         }
10        if (str.equals(rev)) {
11            System.out.println("Palindrome");
12        } else {
13            System.out.println("Not Palindrome");
14        }
15    }
16 }

```

Run

Save

INPUT

```

aaba

```

OUTPUT

```

Palindrome

```

52. Occurrence



SIMATS Online Programming Lab

JAVA PROGRAMMING


K.Dhillip

12/04/2023 10:00

QUESTIONS
11 / 25 completed

Occurrence

OCCURRENCE

Write a Java program to count the number of occurrences of a given character in a string.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         char ch = sc.next().charAt(0);
7         int count = 0;
8         for (int i = 0; i < str.length(); i++) {
9             if (str.charAt(i) == ch) {
10                 count++;
11             }
12         }
13         System.out.println("Occurrences = " + count);
14     }
15 }

```

Run

Save

INPUT

```

java
a

```

OUTPUT

```

Occurrences = 3

```

53. Check



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K. Divya
123456789

QUESTIONS 12 / 20 completed

CHECK
 Write a Java program to check if a given string contains only digits

PROGRAM EDITOR

java
 Run
Save

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         boolean digits = true;
7         for (int i = 0; i < str.length(); i++) {
8             if (!Character.isDigit(str.charAt(i))) {
9                 digits = false;
10                break;
11            }
12        }
13        if (digits)
14            System.out.println("Contains Only Digits");
15        else
16            System.out.println("Does Not Contain Only Digits");
17    }
18 }
            
```

OUTPUT
 Contains Only Digits

INPUT
 123456

54. Length



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K. Divya
123456789

QUESTIONS 12 / 28 completed

LENGTH
 Write a Java program to find the length of the longest substring without repeating characters in a given string.

PROGRAM EDITOR

java
 Run
Save

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         int max = 0;
7         for (int i = 0; i < str.length(); i++) {
8             String temp = "";
9             for (int j = i; j < str.length(); j++) {
10                char ch = str.charAt(j);
11                if (temp.indexOf(ch) != -1)
12                    break;
13                temp = temp + ch;
14                if (temp.length() > max)
15                    max = temp.length();
16            }
17        }
18        System.out.println("Length of Longest Substring = " + max);
19    }
20 }
            
```

OUTPUT
 Length of Longest Substring = 3

INPUT
 abcabcbb

55. Lower



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhrip
15236376140

QUESTIONS 4 / 20 completed

Lower

LOWER

Write a Java program to convert a given string to lowercase.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         String lower = str.toLowerCase();
7         System.out.println("Lowercase String = " + lower);
8     }
9 }
```

OUTPUT

Lowercase String = hello world

INPUT

Hello World

56. Remove



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhrip
15236376140

QUESTIONS 16 / 20 completed

Remove

REMOVE

Write a Java program to remove all white spaces from a given string.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         String result = str.replace(" ", "");
7         System.out.println("String without spaces = " + result);
8     }
9 }
```

OUTPUT

String without spaces = Hello@world

INPUT

Hello World

57. Reverse



SIMATS Online Programming Lab
JAVA PROGRAMMING

K.Dhilly
10/10/2024

QUESTIONS 12 / 20 completed

Reverse

REVERSE
Write a Java program to reverse a given string.

PROGRAM EDITOR

JavaRunSave

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         String reverse = "";
7         for (int i = str.length() - 1; i >= 0; i--){
8             reverse = reverse + str.charAt(i);
9         }
10        System.out.println(reverse);
11    }
12 }
```

OUTPUT

allert

INPUT
Hello

58. Concatenate



SIMATS Online Programming Lab
JAVA PROGRAMMING

K.Dhilly
10/10/2024

QUESTIONS 12 / 20 completed

Concatenate

CONCATENATE
Write a Java program to concatenate two given strings.

PROGRAM EDITOR

JavaRunSave

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String str1=sc.nextLine();
6         String str2=sc.nextLine();
7         String result=str1+str2;
8         System.out.println(result);
9     }
10 }
```

OUTPUT

HelloWorld

INPUT
Hello
World

59. Repeat



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhulip
16/08/2020

QUESTIONS: 12 / 21 completed
Repeat
REPEAT
Write a Java program to find the first non-repeated character in a given string.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String str=sc.nextLine();
6         for (int i=0;i<str.length();i++){
7             char ch=str.charAt(i);
8             int count=0;
9             for (int j=0;j<str.length();j++){
10                 if (ch == str.charAt(j)){
11                     count++;
12                 }
13             }
14             if(count==1){
15                 System.out.println(ch);
16                 break;
17             }
18         }
19     }
20 }

```

OUTPUT

a

INPUT

abccad

60. Frequent



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhulip
16/08/2020

QUESTIONS: 12 / 28 completed
Frequent
FREQUENT
Write a Java program to find the most frequent word in a given string.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         String str=sc.nextLine();
6         String[] words=str.split(" ");
7         String mostFrequent="";
8         int maxCount=0;
9         for (int i=0;i<words.length;i++){
10             int count=0;
11             for (int j=0;j<words.length;j++){
12                 if(words[i].equals(words[j])){
13                     count++;
14                 }
15             }
16             if(count>maxCount){
17                 maxCount=count;
18                 mostFrequent=words[i];
19             }
20         }
21         System.out.println(mostFrequent);

```

OUTPUT

java

INPUT

java is easy and java is powerful

61. Substring



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhillip
19/04/2018

QUESTIONS 12 / 30 completed

Substring

SUBSTRING

Write a Java program to create a substring from a given string.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String str=sc.nextLine();
6         int start=sc.nextInt();
7         int end=sc.nextInt();
8         String result=str.substring(start,end);
9         System.out.println(result);
10    }
11 }

```

OUTPUT

HelloWorld
2
7

62. Compare



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhillip
19/04/2018

QUESTIONS 12 / 30 completed

Compare

COMPARE

Write a Java program to compare two strings lexicographically.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String str1=sc.nextLine();
6         String str2=sc.nextLine();
7         int result=str1.compareTo(str2);
8         if (result==0){
9             System.out.println("Equal");
10        }else if (result<0){
11            System.out.println("First string is smaller");
12        }else{
13            System.out.println("First string is greater");
14        }
15    }
16 }

```

OUTPUT

First string is smaller

63. Logger

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 8 / 8 completed

PROGRAM EDITOR

Java
 Run
Save

INPUT

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class called "Log4jLogger".

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc=new Scanner(System.in);
5         Logger logger=new Log4jLogger();
6         String message=sc.nextLine();
7         logger.debug(message);
8         logger.info(message);
9         logger.warning(message);
10        logger.error(message);
11    }
12 }
13 interface Logger {
14     void debug(String message);
15     void info(String message);
16     void warning(String message);
17     void error(String message);
18 }
19 class Log4jLogger implements Logger{
20     public void debug(String message){
21         System.out.println("DEBUG: "+message);
    
```

OUTPUT

```

DEBUG: Application started
INFO: Application started
WARNING: Application started
ERROR: Application started
    
```

INPUT

Application started

64. Character

SIMATS Online Programming Lab
JAVA PROGRAMMING

QUESTIONS 8 / 8 completed

PROGRAM EDITOR

Java
 Run
Save

INPUT

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of

```

1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         Character c;
6         //System.out.println("Enter 1 for Hero");
7         //System.out.println("Enter 2 for Villain");
8         //System.out.println("Enter 3 for Monster");
9         int choice=sc.nextInt();
10        if (choice==1) {
11            c = new Hero();
12        } else if (choice == 2) {
13            c = new Villain();
14        } else {
15            c = new Monster();
16        }
17        c.move();
18        c.attack();
19        c.takeDamage();
20    }
21 }
    
```

OUTPUT

```

Hero is moving
Hero is attacking
Hero takes damage
    
```

INPUT

1

65. Sort



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhulip
Full Stack Dev

QUESTIONS: 8 / 8 completed

Sort

SORT
Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         TreeSet<Integer> ts=new TreeSet<>();
5         Scanner sc=new Scanner(System.in);
6         int n=sc.nextInt();
7         for(int i=0;i<n;i++){
8             ts.add(sc.nextInt());
9         }
10        System.out.println(ts);
11    }
12 }

```

OUTPUT

[1, 2, 3, 44, 55]

INPUT

5
2
3
1
55
44

66. HashSet



SIMATS Online Programming Lab
JAVA PROGRAMMING

 K.Dhulip
Full Stack Dev

QUESTIONS: 8 / 8 completed

HashSet

HASHSET
Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.HashSet;
2 import java.util.Scanner;
3 public class Main{
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6         HashSet<String> set = new HashSet<>();
7         int n = sc.nextInt();
8         sc.nextLine();
9         for (int i = 0; i < n; i++){
10             set.add(sc.nextLine());
11         }
12         String value = sc.nextLine();
13         set.remove(value);
14         System.out.println("HashSet after removing element: " + set);
15     }
16 }

```

OUTPUT

HashSet after removing element: [Apple, Mango, Orange]

INPUT

4
Apple
Banana
Mango
Orange
Banana

67. TreeMap



SIMATS Online Programming Lab
JAVA PROGRAMMING


 K.Dhilly
NEW PROFILE

QUESTIONS: 8 / 8 completed
 TreeMap
TREEMAP
 Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR
 java

Run
 Save

```

1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         Map<Integer,String> m=new HashMap<>();
6         m.put(101,"Bob");
7         m.put(102,"Sam");
8         m.put(103,"Ram");
9         for(Map.Entry<Integer,String> e:m.entrySet()){
10             System.out.println(e.getKey()+" "+e.getValue());
11         }
12     }
13 }

```

OUTPUT
 101-Bob
 102-Sam
 103-Ram

INPUT
 Your INPUT goes here!

68. PriorityQueue



SIMATS Online Programming Lab
JAVA PROGRAMMING


 K.Dhilly
NEW PROFILE

QUESTIONS: 8 / 8 completed
 PriorityQueue
PRIORITYQUEUE
 Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR
 java

Run
 Save

```

1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         PriorityQueue<Integer> pq=new PriorityQueue<>();
5         Scanner sc=new Scanner(System.in);
6         int r=sc.nextInt();
7         for(int i=0;i<r;i++){
8             pq.add(sc.nextInt());
9         }
10        System.out.println(pq);
11    }
12 }

```

OUTPUT
 [1, 2, 3, 4, 5]

INPUT
 5
 1
 2
 3
 4
 5

69. LinkedHashMap

SIMATS Online Programming Lab
JAVA PROGRAMMING

K. Divya
780 780001 0000

QUESTIONS: 8 / 9 completed

LinkedHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.LinkedHashMap;
2 import java.util.Scanner;
3 public class Main {
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
7         int n = sc.nextInt();
8         for (int i = 0; i < n; i++){
9             String key = sc.next();
10            int value = sc.nextInt();
11            map.put(key, value);
12        }
13        System.out.println("Keys in insertion order:");
14        for (String key : map.keySet()){
15            System.out.println(key);
16        }
17    }
18 }

```

OUTPUT

```

Keys in insertion order:
Apple
Banana
Mango

```

INPUT

```

3 Apple 10 Banana 20
Mango 30

```

70. Max-Min

SIMATS Online Programming Lab
JAVA PROGRAMMING

K. Divya
780 780001 0000

QUESTIONS: 9 / 9 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         Set<Integer> s=new HashSet<>();
6         int n=sc.nextInt();
7         for(int i=0;i<n;i++){
8             s.add(sc.nextInt());
9         }
10        System.out.println("The Added values are : "+s);
11        int min=Collections.min(s);
12        int max=Collections.max(s);
13        System.out.println("Min : "+min);
14        System.out.println("Max : "+max);
15    }
16 }

```

OUTPUT

```

The Added values are : [1, 1001, 44, 332, 12]
Min : 1
Max : 1001

```

INPUT

```

5
44
332
1001
1
12

```